

Topic: Joint Random Variables and Joint Distributions

Definitions

- **Joint PMF (Discrete Case):** For discrete random variables $X \in \{x_1, x_2, \dots, x_n\}$ and $Y \in \{y_1, y_2, \dots, y_m\}$, the joint probability mass function is defined as:

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

This computes the probability of the joint outcome occurring simultaneously.

- **Joint PDF (Continuous Case):** For continuous random variables X and Y , a function $f(x, y)$ is the joint probability density function if the probability that the pair (X, Y) lies in a small rectangle of area $dx dy$ is approximately $f(x, y)dx dy$.

- **Joint CDF:** The joint cumulative distribution function for random variables X and Y is defined as:

$$F(x, y) = P(X \leq x, Y \leq y)$$

Assumptions

- In the **discrete case**, the ordered pair (X, Y) takes values in the product set $\{(x_1, y_1), (x_1, y_2), \dots, (x_n, y_m)\}$.
- In the **continuous case**, if $X \in [a, b]$ and $Y \in [c, d]$, the pair (X, Y) takes values in the product region $[a, b] \times [c, d]$.
- When random variables are not independent, covariance and correlation are used as measures of the nature of the dependence between them.

Key Results

Discrete Joint PMF Properties

1. $0 \leq p(x_i, y_j) \leq 1$.
2. $\sum_{i=1}^n \sum_{j=1}^m p(x_i, y_j) = 1$.

Continuous Joint PDF Properties

1. $f(x, y) \geq 0$.
2. $\int_c^d \int_a^b f(x, y) dx dy = 1$.
3. The joint PDF $f(x, y)$ may take values greater than 1; it is a probability density, not a probability.

Formulas for CDF and PDF Recovery

- **Continuous Joint CDF:** Obtained by integrating the density:

$$F(x, y) = \int_{-\infty}^y \int_{-\infty}^x f(u, v) du dv$$

- **Recovering Joint PDF:** Using partial derivatives:

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$$

- **Discrete Joint CDF:** Obtained by summing probabilities:

$$F(x, y) = \sum_{x_i \leq x} \sum_{y_j \leq y} p(x_i, y_j)$$

Properties of the Joint CDF

- $F(x, y)$ is **non-decreasing**; if x or y increase, then $F(x, y)$ must stay constant or increase.
 - $\lim_{(x, y) \rightarrow (-\infty, -\infty)} F(x, y) = 0$ (Lower-left boundary).
 - $\lim_{(x, y) \rightarrow (\infty, \infty)} F(x, y) = 1$ (Upper-right boundary).
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Examples

Example 1: Rolling Two Dice (Discrete PMF)

Experiment: Roll two fair dice.

Let X = value on the first die and T = total (sum) of both dice.

Each outcome (i, j) has a probability of $\frac{1}{36}$. The resulting joint probability table consists of entries that are either 0 or $\frac{1}{36}$. This establishes that X and T are **not independent**.

Example 2: Joint PDF Validation (Continuous PDF)

Given: $f(x, y) = 4xy$, for $0 \leq x \leq 1$ and $0 \leq y \leq 1$.

Step 1: Check Validity (Normalization)

$$\int_0^1 \int_0^1 4xy \, dx \, dy = \int_0^1 \left(\int_0^1 4xy \, dx \right) dy = \int_0^1 2y \, dy = [y^2]_0^1 = 1$$

Because $f(x, y) \geq 0$ and the total probability equals 1, it is a valid joint PDF.

Step 2: Compute $P(A)$ for event $A = \{X < 0.5, Y > 0.5\}$

$$\begin{aligned} P(A) &= \int_0^{0.5} \int_{0.5}^1 4xy \, dy \, dx = \int_0^{0.5} [2xy^2]_{0.5}^1 dx \\ &= \int_0^{0.5} 2x(1 - 0.25) \, dx = \int_0^{0.5} \frac{3}{2}x \, dx = \left[\frac{3}{4}x^2 \right]_0^{0.5} = \frac{3}{16} \end{aligned}$$